

Save Bandwidth by Setting Up a Fedora



Presenting a step-by-step guide on how to set up a Fedora mirror.

The most common nocturnal activity of an engineering student, particularly when exams are approaching, is to fire up a first-person shooter game. Let's suppose the geek in you, fed up with all the trivialities a textbook has to offer, decides to challenge your friend to a deadly duel of OpenArena. According to Murphy's Law, he won't have it installed, and the Internet connection will go down at that very moment.

Now being a geek, though you are sure that the world is conspiring against you, you won't give up so easily, would you? You will decide to see this through to the end by creating a local mirror of the Fedora Linux distribution, so that every package is ready to serve when you want it. Of course, you will be able to do it yourself, but I consider it my duty to make it easier so as to allow you to take up more important duties like running an OpenArena server.

Now, having read so far, if you are not entirely sure what this is all about, let me tell you: it's about mirroring Fedora repositories within your organisation or institute. The benefits: faster downloads for you and your friends, effective use of

bandwidth and lowered cost.

According to Wikipedia, "In computing, a mirror is an exact copy of a data set. On the Internet, a mirror site is an exact copy of another Internet site." When you try to install a new package into your Fedora installation, either using PackageKit or Yum, it tries to fetch the packages from an Internet site along with the libraries and other software required for it, and install it on your computer. Now software like OpenOffice.org or OpenArena are very big and along with all their dependencies, the download size may be in the order of hundreds of megabytes.

Let's consider a simple calculation; if your organisation has 100 users and each downloads OpenOffice.org separately, it will take around 100×150 MB (which is equal to 15000 MB) of aggregated downloads. If you consider even a normal usage scenario, where users occasionally install new software and update their system, such downloads can easily reach the terabyte levels per month.

In countries like India, where bandwidth is a costly commodity, it is hardly possible for an organisation to invest in an astronomical amount for bandwidth, and this can easily put